UNIVERSITY OF TORONTO

TRANSCRIPT OF CONSOLIDATED ACADEMIC RECORD

TORONTO, ONTARIO, CANADA

NAME: **Agro, Benjamin Taylor Caldwell**

ISSUED TO: Ben Agro

RECORD AS OF: 2025-10-15
STUDENT NUMBER: 1005857020
OEN: 382500007
BIRTH DAY/MONTH: 25/08

REGISTRATION HISTORY

2019 Fall - 2023 Winter : Faculty of Applied Science & Engineering

2023 Fall - 2025 Summer : School of Graduate Studies

GRADUATION SUMMARY

BACHELOR OF APPLIED SCIENCE IN ENGINEERING SCIENCE CONFERRED - JUNE 2023 WITH HIGH HONOURS

Faculty of Applied Science & Engineering

2019 Fall-Dean's Honour List
2020 Winter-Dean's Honour List
2020 Fall-Dean's Honour List
2021 Winter-Dean's Honour List
2021 Fall-Dean's Honour List
2022 Winter-Dean's Honour List
2022 Fall-Dean's Honour List
2023 Winter-Dean's Honour List

2019 FALL - BASC IN ENGINEERING SCIENCE - DIVISION OF ENGINEERING SCIENCE

SESSIONAL GPA: **4.00** CUMULATIVE GPA: **4.00**
SESSIONAL PERS: **98.2**
STATUS: **PASS WITH HONOURS**

CRS CODE	TITLE	WGT	MRK	GRD	CRS AVG
CIV102H1	STRUCT. & MATERIALS	**0.50**	**95**	**A+**	**B**
ESC101H1	PRAXIS I	**0.50**	**98**	**A+**	**B+**
ESC103H1	ENG. MATHEMATICS & COMPUTATION	**0.50**	**100**	**A+**	**B+**
ESC180H1	INTRO TO COMPUTER PROGRAMMING	**0.50**	**98**	**A+**	**B+**
ESC194H1	CALCULUS I	**0.50**	**99**	**A+**	**B**
PHY180H1	CLASSICAL MECHANICS	**0.50**	**99**	**A+**	**B**

May proceed

2020 WINTER - BASC IN ENGINEERING SCIENCE - DIVISION OF ENGINEERING SCIENCE

In the 2019-20 academic year, the University of Toronto was affected by the global COVID-19 pandemic. Instructional methods were modified and some students were graded on the University's approved Credit/No Credit scale for courses completed in Winter 2020.
For more information, see:
http://www.transcripts.utoronto.ca/guide

SESSIONAL GPA: **4.00** ANNUAL GPA: **4.00** CUMULATIVE GPA: **4.00**
SESSIONAL PERS: **96.3**
STATUS: **PASS WITH HONOURS**

CRS CODE	TITLE	WGT	MRK	GRD	CRS AVG
ECE159H1	FUNDAMENTALS OF ELEC. CIRCUITS	**0.50**	**93**	**A+**	**B+**
ESC102H1	PRAXIS II	**0.50**	**96**	**A+**	**B+**
ESC190H1	COMP. ALGOR. & DATA STRUCTURES	**0.50**	**96**	**A+**	**B-**
ESC195H1	CALCULUS II	**0.50**	**96**	**A+**	**B**
MAT185H1	LINEAR ALGEBRA	**0.50**	**98**	**A+**	**B-**
MSE160H1	MOLECULES AND MATERIALS	**0.50**	**99**	**A+**	**B+**

May proceed

2020 FALL - BASC IN ENGINEERING SCIENCE - DIVISION OF ENGINEERING SCIENCE

SESSIONAL GPA: **4.00** CUMULATIVE GPA: **4.00**
SESSIONAL PERS: **96.7**
STATUS: **PASS WITH HONOURS**

CRS CODE	TITLE	WGT	MRK	GRD	CRS AVG
AER210H1	VECTOR CALC. & FLUID MECHANICS	**0.50**	**100**	**A+**	**B**
CHE260H1	THERMODYNAMICS	**0.50**	**98**	**A+**	**B**
ECE253H1	DIGITAL & COMPUT.SYS	**0.50**	**100**	**A+**	**B-**
ESC203H1	ENGINEERING AND SOCIETY	**0.50**	**91**	**A+**	**B**
MAT292H1	ORDINARY DIFFERENTIAL EQUATION	**0.50**	**92**	**A+**	**B**
PHY293H1	WAVES AND MODERN PHYSICS	**0.50**	**99**	**A+**	**B+**

May proceed

2021 WINTER - BASC IN ENGINEERING SCIENCE - DIVISION OF ENGINEERING SCIENCE

SESSIONAL GPA: **4.00** ANNUAL GPA: **4.00** CUMULATIVE GPA: **4.00**
SESSIONAL PERS: **96.2**
STATUS: **PASS WITH HONOURS**

CRS CODE	TITLE	WGT	MRK	GRD	CRS AVG
BME205H1	FUNDAMENTALS OF BIOMEDICAL ENG	**0.50**	**91**	**A+**	**B+**
CHE374H1	ECO.ANA.& DEC.MAKING	**0.50**	**96**	**A+**	**B+**
ECE259H1	ELECTRICITY AND MAGNETISM	**0.50**	**100**	**A+**	**B-**
ECE286H1	PROBABILITY & STATISTICS	**0.50**	**100**	**A+**	**B-**
HPS346H1	Modifying and Optimizing Life	**0.50**	**92**	**A+**	**B+**
PHY294H1	QUANTUM AND THERMAL PHYSICS	**0.50**	**98**	**A+**	**A-**

May proceed

2021 FALL - BASC IN ENGINEERING SCIENCE(MAJOR IN ROBOTICS ENGINEERING) - DIVISION OF ENGINEERING SCIENCE

SESSIONAL GPA: **4.00** CUMULATIVE GPA: **4.00**
SESSIONAL PERS: **97.8**
STATUS: **PASS WITH HONOURS**

CRS CODE	TITLE	WGT	MRK	GRD	CRS AVG
AER301H1	DYNAMICS	**0.50**	**99**	**A+**	**B**
CSC263H1	Data Structures & Analysis	**0.50**	**100**	**A+**	**C+**
ESC204H1	PRAXIS III	**0.50**	**92**	**A+**	**A-**
ESC301H1	ENG. SCI:OPTION SEMINAR	**0.25**		**IPR**	
MIE366H1	Electronics for Robotics	**0.50**	**98**	**A+**	**B**
ROB301H1	INTRODUCTION TO ROBOTICS	**0.50**	**99**	**A+**	**B+**
ROB310H1	MATHEMATICS FOR ROBOTICS	**0.50**	**99**	**A+**	**B+**

May proceed

2022 WINTER - BASC IN ENGINEERING SCIENCE(MAJOR IN ROBOTICS ENGINEERING) - DIVISION OF ENGINEERING SCIENCE

SESSIONAL GPA: **4.00** ANNUAL GPA: **4.00** CUMULATIVE GPA: **4.00**
SESSIONAL PERS: **96.6**
STATUS: **PASS WITH HONOURS**

CRS CODE	TITLE	WGT	MRK	GRD	CRS AVG
AER372H1	CONTROL SYSTEMS	**0.50**	**96**	**A+**	**B-**
CSC384H1	Intro Artif Intell	**0.50**	**98**	**A+**	**B+**
ECE353H1	SYSTEMS SOFTWARE	**0.50**	**97**	**A+**	**B-**
ESC301H1	ENG. SCI:OPTION SEMINAR	**0.25**		**CR**	*****
MIE438H1	MIC. CTRL & EMBED. MIC. PROC	**0.50**	**94**	**A+**	**B+**
ROB313H1	INTRO TO LEARNING FROM DATA	**0.50**	**98**	**A+**	**B+**

May proceed

2022 FALL - BASC IN ENGINEERING SCIENCE(MAJOR IN ROBOTICS ENGINEERING) - DIVISION OF ENGINEERING SCIENCE

SESSIONAL GPA: **4.00** CUMULATIVE GPA: **4.00**
SESSIONAL PERS: **96.8**
STATUS: **PASS WITH HONOURS**

CRS CODE	TITLE	WGT	MRK	GRD	CRS AVG

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ECE470H1	ROBOT MODELING AND CONTROL	**0.50**	**98**	**A+**	**B**
ECE557H1	LINEAR CONTROL THEORY	**0.50**	**100**	**A+**	**B+**
ESC499Y1	THESIS	**1.00**		**IPR**	
JRE410H1	MARKETS & COMPETITIVE STRATEGY	**0.50**	**94**	**A+**	**B+**
ROB501H1	COMPUTER VISION FOR ROBOTICS	**0.50**	**95**	**A+**	**B**

May proceed

2023 WINTER - BASC IN ENGINEERING SCIENCE(MAJOR IN ROBOTICS ENGINEERING) - DIVISION OF ENGINEERING SCIENCE
SESSIONAL GPA: **4.00** ANNUAL GPA: **4.00** CUMULATIVE GPA: **4.00**
SESSIONAL PERS: **98.5**
STATUS: **PASS WITH HONOURS**

CRS CODE	TITLE	WGT	MRK	GRD	CRS AVG
APS511H1	INVENTIONS & PATENTS FOR ENG.	**0.50**	**99**	**A+**	**B+**
CSC412H1	Uncertainty & Learng	**0.50**	**96**	**A+**	**B+**
ESC499Y1	THESIS	**1.00**	**100**	**A+**	**A-**
PER001H1	PRACTICAL EXPER. REQUIREMENT	**0.00**		**CR**	
ROB498H1	ROBOTICS CAPSTONE DESIGN	**0.50**	**96**	**A+**	**A**
ROB521H1	MOBILE ROBOTICS AND PERCEPTION	**0.50**	**100**	**A+**	**B+**

May proceed

School of Graduate Studies

2023 FALL - DOCTOR OF PHILOSOPHY - DEPARTMENT OF COMPUTER SCIENCE

CRS CODE	TITLE	WGT	MRK	GRD	CRS AVG
CSC2420H	Algorithm Design, Analysis and Theory	**0.50**		**A+**	
CSC2508H	Advanced Data Systems	**0.50**		**A**	
RST9999Y	Research/Thesis	**0.00**		**IPR**	

2024 WINTER - DOCTOR OF PHILOSOPHY - DEPARTMENT OF COMPUTER SCIENCE

CRS CODE	TITLE	WGT	MRK	GRD	CRS AVG
CSC2530H	Computer Vision for Advanced Digital Photography	**0.50**		**A+**	
CSC2532H	Statistical Learning Theory	**0.50**		**A+**	
RST9999Y	Research/Thesis	**0.00**		**IPR**	

2024 SUMMER - DOCTOR OF PHILOSOPHY - DEPARTMENT OF COMPUTER SCIENCE

CRS CODE	TITLE	WGT	MRK	GRD	CRS AVG
RST9999Y	Research/Thesis	**0.00**		**IPR**	

2024 FALL - DOCTOR OF PHILOSOPHY - DEPARTMENT OF COMPUTER SCIENCE

CRS CODE	TITLE	WGT	MRK	GRD	CRS AVG
CSC2545H	Advanced Topics in Machine Learning Geometric Deep Learning	**0.50**		**A+**	**EXT**
CSC2626H	Imitation Learning for Robotics	**0.50**		**A+**	**EXT**
RST9999Y	Research/Thesis	**0.00**		**IPR**	

2025 WINTER - DOCTOR OF PHILOSOPHY - DEPARTMENT OF COMPUTER SCIENCE

CRS CODE	TITLE	WGT	MRK	GRD	CRS AVG
RST9999Y	Research/Thesis	**0.00**		**TRF**	

Transfer approved with full credit for work completed in previous program.

2025 SUMMER - MASTER OF SCIENCE - DEPARTMENT OF COMPUTER SCIENCE

CRS CODE	TITLE	WGT	MRK	GRD	CRS AVG
CSC4000Y	Research Project in Computer Science	**1.00**		**CR**	

Angelique Saweczko
University Registrar

End of Transcript

University of Toronto

Guide to Transcript

Effective September 1998

	Undergraduate Grade Scale			Graduate Grade Scale	
Grade Meanings	Refined Letter Grade Scale	Numerical Scale of Marks	Grade Point Value	Truncated Letter Grade Scale	Numerical Scale of Marks
Excellent	A+ A A-	90 – 100% 85 – 89% 80 – 84%	4.0 4.0 3.7	A+ A A-	90 – 100% 85 – 89% 80 – 84%
Good	B+ B B-	77 – 79% 73 – 76% 70 – 72%	3.3 3.0 2.7	B+ B B-	77 – 79% 73 – 76% 70 – 72%
Adequate	C+ C C-	67 – 69% 63 – 66% 60 – 62%	2.3 2.0 1.7		
Marginal	D+ D D-	57 – 59% 53 – 56% 50 – 52%	1.3 1.0 0.7		
Inadequate	F	0 – 49%	0.0	FZ	0 – 69%

Approved grade scales that are outside the standard system:
A passing mark in the Faculty of Dentistry, Faculty of Medicine, and the Faculty of Nursing is 60%.
H (Honours), **P** (Pass), **FL** (Failure).
HH (High Honours), **H** (Honours), **P** (Pass), **LP** (Low Pass), **F** (Fail)
The Faculty of Law J.D. program used the following grade scale between Fall-1998 and Fall-2012:
A:80-100 (Excellent); B+:75-79 (Good); B:70-74 (Good); C+:65-69 (Adequate);
C:60-64 (Adequate); D:50-59 (Marginal); F:below 50 (Inadequate).
In cases where the FL grade is used in calculating grade point averages, the symbol FL% is used and a grade point value of 0.0 is assigned.
CR (Credit), **NCR** (No Credit).
In cases where the NCR grade is used in calculating grade point averages, the symbol NC% is used and a grade point value of 0.0 is assigned.

Other Symbols and Notations:

ADD Additional course, taken for credit. Included in grade point averages.
(note: the following symbols are not used in calculating grade point averages.)
AEG Aegrotat standing granted on the basis of term work and medical or similar evidence. Course credit granted.
DNW Did not write/did not attend/did little work.
EXT Extra course, not for credit.
GWR Grade withheld pending review.
INC Incomplete.
IPR In progress.
(IPR will also appear if the course has not yet started or the course has finished but an approved grade is not yet available.)
LWD Late withdrawal without academic penalty.
NGA No grade available.
SDF Standing deferred on the basis of incomplete course work because of medical or similar reasons.
WDR Withdrawn without academic penalty.
XMP Exemption granted on the basis of credit for work done elsewhere. Course credit granted.
Candidacy Achieved
A student who has completed all program requirements for the PhD degree exclusive of thesis research will be considered a candidate in the School of Graduate Studies.

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Averages Information:

The following types of averages may appear:

GPA Grade Point Average*
(*for CTEP programs, the GPA does not include performance in B.Ed. Courses)

ACCA GPA on the Academic Activity Component of the Bachelor of Physical Health and Education

CICA GPA on the Curriculum and Instruction Component of the Bachelor of Education and Diploma in Technical Education

CTPA GPA on the Bachelor of Education Activity Component of the Concurrent Teacher Education Program

FNCA GPA on Foundations Component of the Bachelor of Education and Diploma in Technical Education

PACA GPA on the Physical Activity Component of the Bachelor of Physical and Health Education

PERS Percentage Average

ACCUM. AVG
Accumulated Average

Awards Information:
Academic awards information is not listed on University of Toronto transcripts.

Academic Sessions:
Fall Session (September – December)
Winter Session (January – April)
Summer Session (May – August)

Course Credits:
A course with a weight of 1.00 equals 6 semester hours.

Course Average:
Course average is shown for undergraduate courses with a size of 12 or more and is expressed as a letter grade correspondingto the refined letter grade scale. The symbol "*" is used to indicate an undergraduate course with fewer than 12 students. A"blank" is used to indicate that a course average has not been calculated.

End of Guide

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